MSc Finance 20127

Corporate Finance (Business Valuation)
03 — Business Forecasting for Valuation

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Where We Are in the Valuation Workflow

Equity story $\longrightarrow$ Financial statements $\longrightarrow$ **Forecasts** $\longrightarrow$ Cash flows $\longrightarrow$ Discount rate $\longrightarrow$
Terminal value $\longrightarrow$ Multiples $\longrightarrow$ Value bridge $\longrightarrow$ Recommendation

Today:
Equity story + financial statements $\longrightarrow$ coherent forecasts $\longrightarrow$ projected cash flows

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Today's Professional Task

Professional task:

- You receive a management business plan
- Your job is ***not*** to copy it into Excel
- Your job is to assess whether it is coherent enough to value
  - identify the business story
  - translate the story into value drivers
  - build integrated forecast statements
  - test whether sales, margins, CapEx, working capital, & financing assumptions fit together
  - understand how JJ Jones' value drivers generate projected statements

Plan for Today

Layer	What We Learn	Why It Matters
CORE	Equity story $\rightarrow$ value drivers $\rightarrow$ forecast statements	JJ Jones, project
CORE	Projecting IS, BS, and CF statements	Integrated valuation model
PRO	Market analysis, strategy, actions, assumptions	Professional business planning
TECH	Model architecture, cash sweep, circularity	Avoid spreadsheet/model errors
RESEARCH PREVIEW	Forecast coherence	Bridge to special topic
PRACTICE	JJ Jones	Turn assumptions into statements

From Equity Story to Value Drivers

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From Story to Value Drivers: Overview I

Business planning consists of several interrelated phases:

1. Thorough analysis of company features, market(s), positioning and past performance
2. Definition of precise competitive strategies
3. Definition of actions needed to pursue these strategies
4. Definition of quantitative assumptions underpinning forecast financial statements
5. Preparation of forecast financial statements

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From Story to Value Drivers: Overview II

- There are at least three significant dangers when numbers are used without any narrative in constructing valuations:



- In a good valuation, numbers are bound together coherently and story-telling is kept grounded with numbers
  - Develop a narrative for the business you are valuing or considering investing in
  - Test narrative against history, experience and common sense
  - Convert key parts of narrative into value drivers
  - Connect value drivers to a valuation
  - Keep the feedback loop open

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Key message: A business plan must be both narratively credible and financially coherent

What is a Coherent Plan?

CORE

A plan is coherent when:

1. the business story explains the value drivers
2. value drivers map into financial-statement items
3. sales growth is supported by capacity, CapEx, and working capital
4. margin improvement has an operational explanation
5. financing assumptions are consistent with cash-flow deficits or surpluses
6. long-run assumptions are reachable from the explicit forecast period

Bottom line: A plan can be plausible line by line and still be incoherent as a system

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JJ Jones: Today's Running Case

We will translate a compact set of value drivers into:

- income statement forecasts
- PPE forecasts
- balance sheet forecasts
- free cash-flow forecasts

From Story to Value Drivers: Market Analysis

PRO

- To prepare accurate and coherent forecasts, it is necessary to analyse the markets the firm operates in
- The articulation and complexity of the analysis depends on several variables:

Unrelated businesses
The firm operates in unrelated businesses (*holding firms*)

↓

Different geographical markets
The firm operates in different geographical markets (*multinational firms*)

↓

Different distribution channels
The business units target different distribution channels(1)

↓

Residual
Hybrid situations that mix the cases above

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(1): For instance, a firm could sell its products through both its proprietary website and a network of retail stores

From Story to Value Drivers: Competitive Strategy

PRO

The competitive strategies should be prepared both at:

1. The overall company level; and
2. Any single business unit level (if any)



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From Story to Value Drivers: Competitive Strategy Example

PRO

Distribution

Increase proximity to target customers

Wholesale	• Direct model in key markets • Trade marketing and space management • Moderate doors increase
Retail	• Retail 2.0 • Network development
eCommerce	• Increase conversion • Develop omnichannel • Boost traffic
B2B	• Large deals development

Source: [redacted] Analysis, management data

Product portfolio

Develop paper and non-paper categories

Paper	• Increase limited editions, premium and personalized products
Bags & SLG	• Reinforce capabilities and penetrate specialized channel
[redacted]	• Expand mainly through franchising (travel) and selective JVs (urban)
Digital	• Development of new products and services with partners
Writing tools	• Evaluate potential tactical partnership to enlarge product range

Current development status: High Low

From Story to Value Drivers: Action-Taking

CORE

- Once defined the competitive strategies, the focus is on the actions needed to implement them(1)
- Each action should be matched with the following prescriptions:



(1): The credibility of a business plan largely depends on how clearly the mapping of the actions to implement it is set out

From Story to Value Drivers: Action-Taking Example

PRO

Initiatives	Key actions
Moderate doors increase	• Expand presence in APAC, selectively increase doors, explore new channels in US (eg. Walmart) to balance decline of bookstores and stationery shops
Direct model in key markets	• Take direct control on additional high potential retailers to gain higher profitability (focus on UK, France, Japan)
Trade marketing and space management	• Increase Visual merchandizing in top-selling doors (gold, platinum) with external agency to increase rotations with in-store activities, enhance space quality (eg. from ground spinner to wall or wall to atelier)

From Story to Value Drivers: Quantitative Assumptions

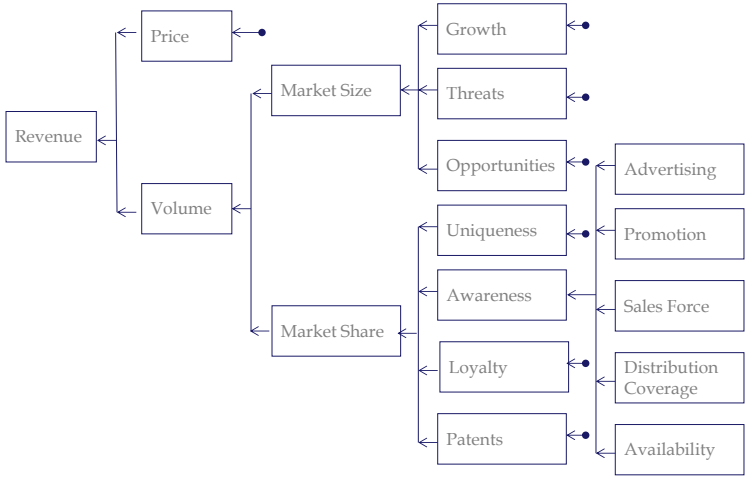
CORE

- The planning activity ends with a forecast financial statement
- The development of such forecast is based on a series of quantitative assumptions

Macro - economic	• Different for each *geographic area* (GDP, inflation, forex, …)
Business Activity	• Evolution of *key financials* (sales, costs, working capital, ratios, …)
Corporate Structure	• Changes at *corporate level* (capital structure, fiscal policy, surplus assets, …)

From Story to Value Drivers: Revenue Forecast Assumptions

CORE



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Professional rule: do not forecast revenue growth as one number unless you can explain the whole logic behind it

From Story to Value Drivers: Modelling

CORE



Integrated Forecast Financial Statement

(This is where the equity story becomes numbers)

Building the Integrated Forecast Model

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Forecasting: Modelling Basics I

CORE

Key steps to create a forecasting model:

1. Set up the model template
2. Input historical financials
3. Calculate ratios
4. Build assumptions
5. Project the IS (leave out interest)

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Forecasting: Main Aspects of Modelling Basics

PRO

Setting up a proper model template, the crucial aspects are:

1. One sheet vs multi-sheet model
2. Matrix integrity
   - A given column refers to a specific year in all sheets
3. Key sheets
   - Assumptions/inputs
   - Secondary calculations
   - Debt schedule
   - Income statement
   - Balance sheet
   - Cash flow statement
   - Summary ratios
   - Output

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Forecasting: Modelling Basics II

CORE

Key steps to create a forecasting model:

6. Project the BS (leave out cash and debt)
7. Project the CF statement
8. Balance the BS
9. Model interest expense/income
10. Create summary/output pages

Forecasting: Main Precautions of Modelling Basics

PRO

Setting up a proper model template you should take the following precautions:

1. Modelling standards
   (a) **Accurate**: always proof read your model (sanity, matrix, and stress check)
   (b) **Coherent**: assumptions about individual items should fit with each other
   - Flexible: no hard numbers in formulae
   - User friendly: break down complex calculations
2. Learn keyboard shortcuts
   - More accurate, more coherent (and more sleep)
3. Use comments to indicate sources
4. Formatting and color coding
   - Write inputs, links to other sheets, and formulae in three different colors (style sheet)
5. Be able to print your model any time

Forecasting: Modelling Basics III

CORE

Collecting historical financial inputs, keep in mind that:

1. A standard model would typically have 3-5 years of historical data
2. Line items for IS and BS should reflect the structure of the company's published financials
3. Some line items need to be forecasted separately even if not broken down in the financials
   - D&A: generally included in COGS and SG&A. Find the total amount in the CF statement
   - Interest expense: can be expressed net of interest income
   - Interest income: can be netted from interest expense or included in other income
   - Non-recurring items: identify and separate them from the remaining (ordinary) line items
4. Some line items might be aggregated for simplicity (other assets/liabilities)

Forecasting: Modelling Basics IV

PRO

In ratio-calculations and assumption-building, you should:

1. Identify key ratios to analyze historical performance:
   - IS (growth/margins)
   - BS (days receivable, payable, inventory)
2. Make assumptions on the evolution of key ratios based on:
   - Trends in historical ratios
   - Strategic assessment
   - Market consensus for the company being modelled
   - Market consensus for competitors (peer group)
3. Cross-check assumptions versus:
   - Historical trends (are there any abrupt jumps?)
   - Internal coherence (is the evolution of related items coherent?)
   - External consistency (is the evolution in line with consensus?)

Forecasting: Modelling Basics V

TECH

Income Statement
Revenues
Operating Expenses
EBIT
Interest Expense
EBT
Taxes
Net Income

Balance Sheet
Operating Assets
Fixed Assets
Total Assets
Operating Liabilities
Debt
Total Liabilities
Preferred Stock
Common Stock
Retained Earnings
Stockholders' Equity
Total Liabilities + Equity

Retained Earnings
Beginning Balance
Net Income
Preferred Stock Dividends
Common Dividends
Ending Balance

Free Cash Flows
EBIT
Taxes on EBIT
Non-Cash Expenses
Change in Required Cash
Change in Working Cap
CAPEX
Unlevered FCF
Interest Expense
Interest Tax Shield
Preferred Dividends
Change in Debt
Change in Preferred Stock
Equity FCF
Common Dividends
Change in Excess Cash
Change in Required Cash
Change in Cash

Supporting Schedules

Competitive Analysis
General Economy
Industry
Comparables

Fixed Assets
Capacity Needs
Fixed Asset Requirements
Capital Expenditures
Depreciation

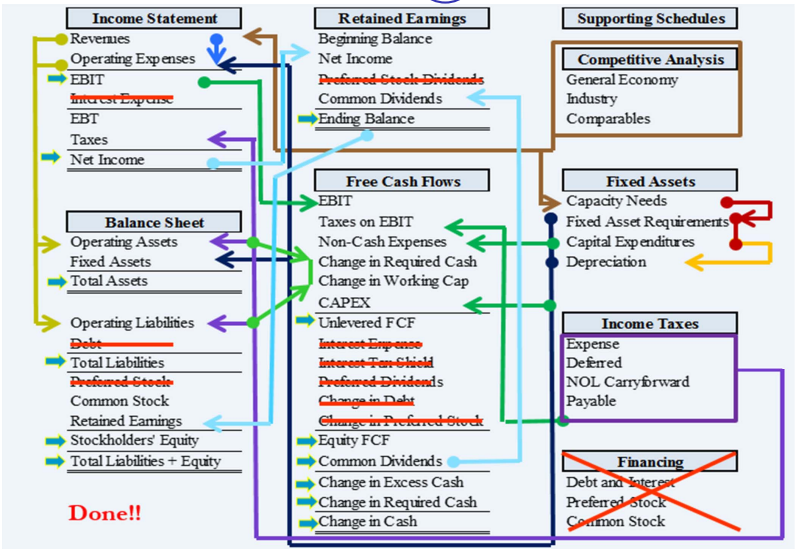
Income Taxes
Expense
Deferred
NOL Carryforward
Payable

Financing
Debt and Interest
Preferred Stock
Common Stock

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Forecasting: Modelling Basics VI



Forecasting: Projecting the Income Statement (IS) I

CORE

You can model the main IS line items as follows:

Sales (*forecast growth*)

Price growth drivers (inflation, strategy)
Volume growth drivers (tot. market *vs* market share)

COGS and SG&A

Project COGS and SG&A as a % of sales

Depreciation

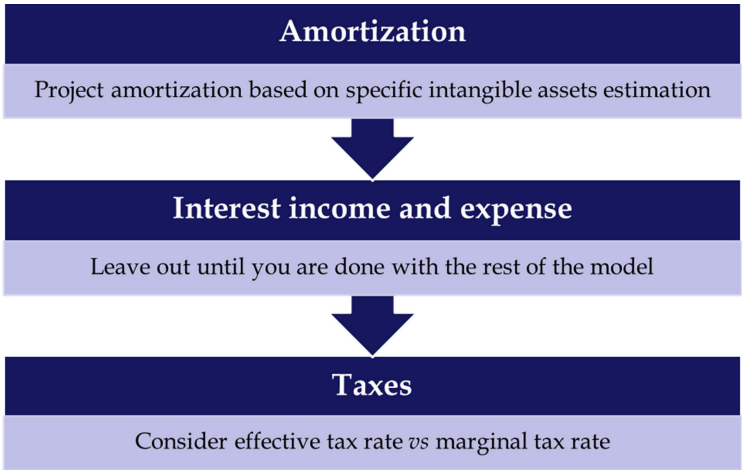
Project depreciation as % of PPE at beginning of the period

Coherence check: If revenue grows, do margins, D&A, and capacity assumptions still make sense?

Forecasting: Projecting the Income Statement (IS) II

CORE

You can model the main IS line items as follows:



Coherence check: If debt, cash, taxes, or interest change, do projected interest and tax assumptions follow?

Forecasting: Projecting the Income Statement (IS) III

PRO

Further issues to consider while projecting the IS:

Net income	• Normalize net income for *non* recurring items
EPS	• Basic *vs* diluted • Net income *vs* normalized net income
DPS	• DPS growth rate *vs* payout ratio
Equity	• Use net income and dividends to complete equity base analysis

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Forecasting: Projecting the Balance Sheet (BS) I

CORE

You can model the main BS line items as follows:

Working capital

Driven by sales and COGS (receivable, inventory and payable days)

Fixed assets ***(use PPE schedule in secondary calcs)***

CAPEX as a fixed component + % of sales. Maintenance *vs* expansion

Goodwill

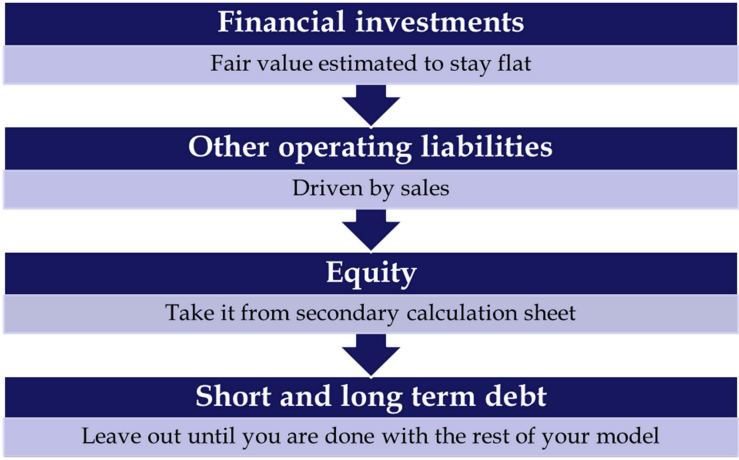
Keep it flat

Coherence check: If revenue grows, do receivables, inventories, payables, and PPE grow consistently?

Forecasting: Projecting the Balance Sheet (BS) II

PRO

You can model the main BS line items as follows:



Coherence check: Are cash and debt balancing the model, or are they hiding operating-forecast mistakes?

Forecasting: Projecting the Cash Flow Statement (CF) I

CORE

- You are 100% certain to be able to obtain the firm's change in cash from the change in all non-cash BS accounts: read that again, you have to be no less than 100% certain
- CF captures changes in every balance sheet items and yields resulting change in cash
- Some general rules in how the CF statement works:
  - Assets up, cash down
  - Liabilities up, cash up
  - Change in WC captures changes in operating assets/liabilities
  - CAPEX and D&A capture changes in fixed assets
  - Net income and dividend capture changes in retained earnings

Forecasting: Projecting the Cash Flow Statement (CF) II

PRO

1. Net Cash Flow
   - \+ Operating cash flow
   - \+ Investing cash flow
   - \+ Financing cash flow
   - = Net Cash Flow (i.e., Δ cash)

$$\text{Beginning Cash} + \text{Net Cash Flow} = \text{Ending Cash}$$

2. Cash Available for Debt Repayment (CADR)
   - \+ Operating cash flow (Starting from NI)
   - \+ Investing cash flow
   - \+ Equity issuance
   - – Dividends/stock repurchase
   - = Cash Flow Available for Debt Repayment (CFADR)
   - – Cash Required (for day-to-day activities)
   - = Cash Available for Debt Repayment (CADR)

$$\text{Beginning Cash} + \text{CFADR} - \text{Cash Required} = \text{CADR}$$

Forecasting: Final Procedures I

TECH

- In order to balance the BS of your model, you need to:
  - Complete the debt schedule
  - In the exercise, we use a simple **cash sweep**(1):
    - If cash available for debt repayment is positive, it goes to repay debt
    - If cash available for debt repayment is negative, short term debt is issued to cover the shortage of cash and the minimum level of operating cash required to run operations smoothly
  - Link resulting:
    - Short term, and
    - Long term debt balance to BS (in turn this is reflected in the CF statement)
  - Link the resulting cash balance in the CF statement to the BS

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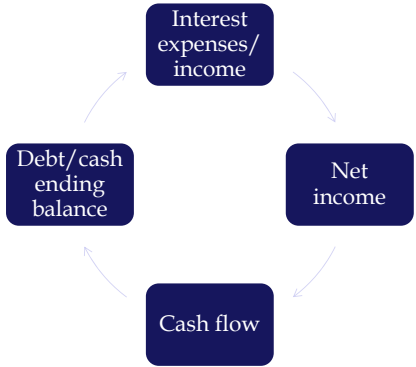
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(1): All cash available for debt repayment is used to pay down debt starting from short and moving to long term debt

Forecasting: Final Procedures II

TECH

- When interest expense is computed on average debt and interest income on average cash, the model becomes circular



- A (less accurate) alternative that does not require iteration is to use opening balance for debt and cash

Forecasting: Final Procedures III

PRO

- Lastly, we create a summary, and/or output pages to present the model
- In this respect, we need to:
  - Build ratios on projected performance:
    - Growth
    - Activity/asset turnover
    - Solvency/credit
    - Profitability
    - Returns
  - If needed, prepare output sheets:
    - Easy to copy-paste into presentation (life saver when last minute change to assumptions, e.g., before board meeting starts)
    - Clear and readable when you print directly from Excel

JJ Jones Case: From Value Drivers to Forecast Statements

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JJ Jones: Value Drivers

Income Statement

- Revenue growth — simple 10% growth
- Cost of goods sold — 60% of revenues
- Operating expenses — 10% of revenues
- Depreciation — 10 years
- Income tax rate — 40% on all income

Assets

- Excess assets (cash and other) — $0
- Required cash balance — 5% of revenues
- Accounts receivable — 50 days*
- Inventory — 80 days*
- Other current assets — 6% of revenues
- Net PPE to revenues — 1.2 ratio*
- Other assets — $0

Liabilities and Preferred Stock

- Accounts payable — 40 days*
- Accrued expenses — 20% of revenues
- Non-current liabilities — 10% of revenues
- Other income — $0 (excess assets related)
- Interest rate — 8% (beginning balance)
- Preferred dividend rate — 10% (beginning balance)
- Debt — retire current portion of long-term debt, increase long-term if the company has insufficient cash flow (same interest rate)
- Preferred stock — no change

* based on ending balance (not average balance)

Discussion Questions

Given the value drivers for JJ Jones of the previous slide, prepare:

i. JJ Jones' Income Statement Forecasts

ii. JJ Jones' PPE Forecasts

iii. JJ Jones' Balance Sheet Forecasts

iv. JJ Jones' Free Cash Flows Forecasts

What JJ Jones Tests

What JJ Jones tests

1. Can you translate assumptions into forecast statements?
2. Can you keep the three statements linked?
3. Can you distinguish required cash from excess cash?
4. Can you forecast working capital from days ratios?
5. Can you forecast PPE from the PPE/revenue ratio?
6. Can you let debt/cash balance the model instead of plugging operations?
7. Can you compute FCFO and FCFE from the resulting statements?

Common Mistakes in Business Forecasting

1. Forecasting a hockey stick without explaining the operational mechanism
2. Forecasting the most likely case instead of expected cash flows
3. Growing sales while leaving CapEx or working capital flat
4. Improving margins without explaining pricing power, cost efficiency, or mix
5. Ignoring cyclicality and normalizing the wrong year
6. Letting cash or debt become the hidden plug
7. Treating business plan as spreadsheet exercise rather than story-to-numbers exercise

Takeaways

1. A business plan starts with a story, but it is only useful for valuation when the story becomes value drivers
2. Forecasts must be integrated: IS, BS, CF statement, debt schedule, and outputs must reconcile
3. Revenue growth should be decomposed into price, volume, market size, and market share whenever possible
4. CapEx and working capital assumptions must be coherent with sales growth and capacity needs
5. A good model does not hide problems in cash, debt, or plug variables
6. JJ Jones is our first full exercise in translating value drivers into forecast statements and cash flows